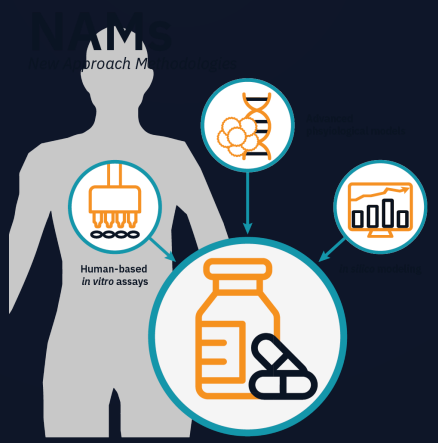
ONDERZOEKSVOORSTEL PITCH · INNOVATIVE TESTING

Multimodal & Multiscale AI for Toxicology

1. NEW APPROACH METHODOLOGIES (NAM'S)



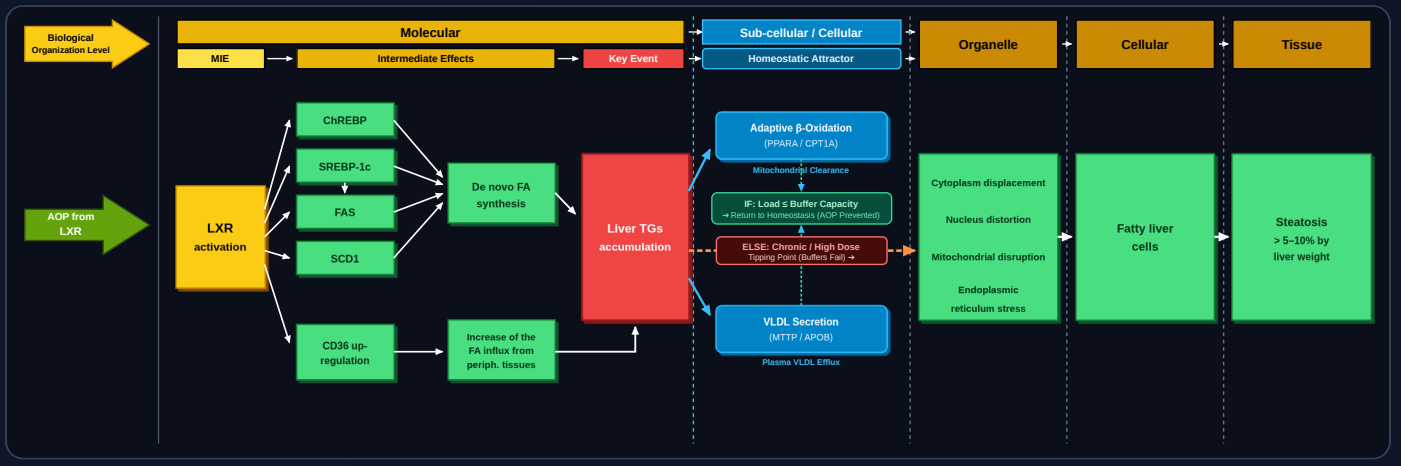
- **Probabilistic Chemical Risk Assessment**
  Accelerating the transition to human-relevant, animal-free biomedical innovation
- **Pillars: (q)AOPs & Physiological Maps**
  Mechanistically mapping and explaining *Adverse Outcomes*

2. EXPANDING ON LINEAR AOPS

This research will deliver a toolbox for the development of quantitative AOP's. From linear chains to dynamic, quantitative systems predicting

$$>P(z_{homeo}, z_{adversity} \mid \text{context, exposure, dose})$$

AOP 34 LXR (MIE 167) → lipogenese → TG-accumulation (KE 291) → steatose



CORTICAL SPREADING DEPRESSION



Breakthrough (left) vs. Propagation Failure (right)
Same MIE: tissue structure determines the fate

SAME MIE, DIFFERENT OUTCOMES

Both tissues initiate with an identical central potassium perturbation (MIE). The outcome arises through spatiotemporal emergence: the intrinsic clearance capacity of the astrocytic syncytium (V_{max}) dictates whether the tissue quenches the local surge or tips into an all-or-none propagating wave.

scan QR for live simulation

Live Simulator

Homeostasis or Adversity?

A DEMO CASE a **multimodal & multiscale model** that fuses chemical, molecular and physiological data into one representation — and turns it into **probabilities**, not a label.

FIVE DATA SOURCES

Chemical information
SMILES · 2D OGB graph + SELFIES

12 compounds · 3 MIE classes

Protein–protein interactions
STRING-db v12 · taxon 9606 · PPI

18 genes · mean score 0.81

Gene expression
GTEx v8 · HPA · GEO GSE126848 · HepaRG

18 genes × 3 states

Gene–gene interactions
OECD AOP 34 & 27 + Physiological Maps

20 edges · 14 + / 6 − · [19,19]

Physiological measurements
MEA · multi-electrode array · in vitro

Multi-modal & Multi-scale AI

Graphormer 47.7M + ToxFormer 25.4M

+ final classification layer (50/50 fusion)

e.g. (N)ODEs

Predicted MIE from 12 compounds

structure-only classifier · held-out LOOCV

MIE 167 · LXRα **MIE 41 · BSEP**

50/50 consensus
leave-one-out
10/12 correct
mean P 0.52
range 0.16–0.67
uncertainty is measurable →

	no MIE	167	41
no MIE	3	0	1
167	0	4	0
41	0	1	3

rows = true · cols = 50/50 LOOCV

$P(z_{homeo}, z_{adversity} \mid \text{context, exposure, dose})$

Probability that the hepatocyte tips to $z_{adversity}$ instead of returning to z_{homeo}, given context, exposure and dose

z_{homeo} $z_{steatosis}$ $z_{cholestasis}$

LEGEND
- done
- possible future work

Sources: OECD AOP 34 · OECD AOP 27 · STRING-db v12 · GTEx v8 / HPA · GEO GSE126848 · HepaRG (Rodrigues 2018)

6 open sources · 18 genes · 12 compounds · 73.1M parameters